

SHENGdong LU

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RESEARCH SUMMARY

I am currently a Postdoctoral Research Associate at Durham University and a member of the COLIBRE collaboration. My research focuses on galaxy formation and evolution, especially the interpretation of JWST observations and the forward modelling of high-redshift galaxy populations using cosmological simulations, semi-analytic models, and radiative transfer.

EMPLOYMENT

Durham University

Postdoctoral Research Associate
Supervisor: Prof. Carlos S. Frenk

Durham, UK
Jul. 2023 – present

Tsinghua University

Postdoctoral Researcher & Research Assistant
Supervisor: Prof. Dandan Xu & Prof. Shude Mao

Beijing, China
Oct. 2020 – Jul. 2023

EDUCATION

National Astronomical Observatories, University of Chinese Academy of Sciences Beijing, China
Ph.D. in Astrophysics
Thesis: *Dynamical Scaling Relations in Observed and Simulated Galaxies*
Supervisor: Prof. Shude Mao
Sept. 2016 – Sept. 2020

University of Science and Technology of China

B.Sc. in Astronomy

Anhui, China
Sept. 2012 – Jul. 2016

RESEARCH THEMES

- High-redshift galaxy formation and evolution, UV luminosity functions
- Cosmological hydrodynamical simulations, semi-analytic models, and radiative-transfer forward modelling
- Interpretation of JWST observations and observable-space inference effects
- Galaxy quenching, circumgalactic angular momentum, and environmental effects
- Galaxy dynamics, stellar populations, and IFU surveys at low redshift
- AI for astronomy

INVITED AND CONTRIBUTED TALKS

- **28th Guoshoujing Meeting on Galaxy Formation and Evolution**, Haikou, China, May 2026, contributed talk
- **SKIRT Days 2025**, Ghent, Belgium, Oct. 2025, contributed talk
- **IAU Symposium #369: Massive Galaxies across the Universe**, Naples, Italy, June 2025, contributed talk

- **COLIBRE Collaboration Meeting 2025**, University of Leiden, Leiden, Netherlands, April 2025, contributed talk
- **Virgo Collaboration Meeting 2025**, University of Sussex, Brighton, UK, Jan. 2025, contributed talk
- **Decades of EXtragalactic (DEX) Workshops – XXI**, Newcastle University, Newcastle upon Tyne, UK, Jan. 2025, contributed talk
- **European Astronomical Society 2024 Annual Meeting**, Padova, Italy, July 2024, poster with short talk
- **Galaxy Evolution Seminar**, University of Oxford, Oxford, UK, Feb. 2024, invited talk
- **24th Guoshoujing Meeting on Galaxy Formation and Evolution**, Huangshan, China, May 2023, contributed talk and session chair
- **Galactic Dynamics Workshop of China**, online, Oct. 2022, organizer
- **23rd Guoshoujing Meeting on Galaxy Formation and Evolution**, Hangzhou, China, May 2021, contributed talk
- **MaNGA Science Conference**, online, July 2020, contributed talk
- **10th C-MaNGA workshop**, Tsinghua University, Beijing, China, Dec. 2019, contributed talk and session chair
- **2019 SDSS-IV: MaNGA Collaboration Meeting**, University of Oxford, Oxford, UK, April 2019, poster with short talk
- **2018 IllustrisTNG Science Workshop**, MPA, Garching, Germany, Oct. 2018, contributed talk

TEACHING

- Machine Learning Demonstrations for MISCADA (MSc in Scientific Computing and Data Analysis), Teaching Assistant and Guest Lecturer, Durham University, 2025 - 2026
- Tutorials on Foundations of Physics, Durham University, 2023 - 2025

MENTORING AND SUPERVISION

- Co-mentored Dr. Sen Wang, then a PhD student at Tsinghua University, together with Prof. Dandan Xu during my postdoctoral appointment at Tsinghua University.
- Co-mentored Mr. Wenyu Zhong, PhD student at Xiamen University, together with his supervisor Prof. Min Du.

PROFESSIONAL SERVICE AND ACTIVITIES

- Referee for MNRAS, A&A, and PASP
- Organizer, Galactic Dynamics Workshop, 2022
- Organizer, Weekly Lunch Talk, Department of Astronomy, Tsinghua University
- Organizer, group journal club discussions

OUTREACH

- Volunteer, The Schools' Science Festival, Durham University, April 2026
- Volunteer, Ukrainian Summer School, Aug. 2024